Guide to Install MATLAB and Simulink for Ubuntu

Specific guide for WingLoop use, March 2026

0.0) Introduction

This guide was written by Leonardo AVONI (avonileonardo@gmail.com) to ease the install procedure of MATLAB and Simulink, to be used within the WingLoop framework. Have fun!

1.0) MATLAB Setup

1.1) Download and Install MATLAB

To access MATLAB zip file:
https://fr.mathworks.com/help/install/ug/install-products-with-internet-connection.html
Basically:

1. Download the Matlab ubuntu zip
2. Unzip it using the following commands (and move to the folder):

```
unzip matlab_R2025b_Linux.zip -d ./matlab_R2025b_Linux
cd ./matlab_R2025b_Linux
```

3. Launch the installation procedure

```
xhost +SI:localuser:root
sudo -H ./install
xhost -SI:localuser:root
```

1.2) Create a MATLAB Desktop Entry

Open a terminal and run the following command. This creates and opens with nano an empty `matlab.desktop` file.

```
sudo nano /usr/share/applications/matlab.desktop
```

Paste this into the `matlab.desktop` file:

```
[Desktop Entry]
Type=Application
Name=MATLAB R2025b
Icon=/usr/local/MATLAB/R2025b/extern/icons/matlab.png
Exec=/usr/local/bin/matlab_open.sh %f
Comment=MATLAB - Numerical Computing Environment
Categories=Development;Science;Math;
Terminal=false
StartupNotify=true
MimeType=text/x-matlab;
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab.desktop` entry executable:

```
sudo chmod +x /usr/share/applications/matlab.desktop
```

1.3) Create a MATLAB Launcher File

Type the following (mkdir the bin directory if it does not exist). This creates an empty `matlab_open.sh` file

```
nano /usr/local/bin/matlab_open.sh
```

Paste the following in that file

```
#!/bin/bash

FILE="$1"

if [ -z "$FILE" ]; then
    /usr/local/MATLAB/R2025b/bin/matlab -desktop
else
    DIR="$(dirname "$FILE")"
    BASE="$(basename "$FILE")"
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "cd('$DIR');
edit('$BASE');"
fi
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab_open.sh` file executable:

```
chmod +x /usr/local/bin/matlab_open.sh
```

When done, you need to refresh the applications database. To do so, either type the following:

```
sudo update-desktop-database
```

Or simply log out and log back in.

Now you should see **MATLAB R2025b** in your Applications menu with an icon.

1.3) Create a MATLAB Icon

The installer normally does not take care of the MATLAB icon, hence

1. Create (or find somewhere) an icon called `matlab.png`
2. Create (if not already there) the icons folder in the MATLAB folder:

```
sudo mkdir /usr/local/MATLAB/R2025b/extern/icons/
```

3. Move the `matlab.png` icon to that folder:

```
sudo mv matlab.png /usr/local/MATLAB/R2025b/extern/icons/
```

1.3) Set-Up the MATLAB Python Engine

NOTE: Only proceed with this part if you already set-up a proper Python environment.

To install the MATLAB-Python API; Matlab specifies some installation procedures:

- https://fr.mathworks.com/help/matlab/matlab_external/install-the-matlab-engine-for-python.html

If you follow those + chatgpt and try to install it using the default install file, it will work (maybe) but there may be issues. At the moment, it seems that the link is available for the python 3.12.x versions; 3.13.x do not work; so make sure you install it in an environment that runs 3.12.x

We have found the following command to work (installs directly from online the latest version). Moreover, it's more straight-forward. The only problemo is that it still (at the moment, March 2026) only works for Python 3.12.x ("Python 3.9, 3.10, 3.11, or 3.12 ") but it's ok

```
python -m pip install matlabengine
```

1.4) Removing old MATLAB or Python Links

When typing “matlab” or “simulink” in the terminal, it is possible that Ubuntu does not understand where to look for the app. It is possible that the following files exist in the `~/.local/share/applications/` folder:

```
mw-matlabconnector.desktop
mw-matlab.desktop
mw-simulink.desktop
```

If so, remove them:

```
rm ~/.local/share/applications/mw-matlabconnector.desktop
rm ~/.local/share/applications/mw-matlab.desktop
rm ~/.local/share/applications/mw-simulink.desktop
```

2.0) Simulink Setup

2.1) Create a Simulink Desktop Entry

Open a terminal and run the following command. This creates and opens with nano an empty `simulink-r2025b.desktop` file.

```
sudo nano /usr/share/applications/simulink-r2025b.desktop
```

Paste this into the `simulink-r2025b.desktop` file:

```
[Desktop Entry]
Type=Application
Name=Simulink R2025b
Comment=Simulink - Model-Based Design Environment
Exec=/usr/local/bin/simulink_open.sh %f
Icon=/usr/local/MATLAB/R2025b/extern/icons/simulink.png
Terminal=false
```

```
StartupNotify=true
Categories=Development;Science;Math;
MimeType=application/x-simulink;
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `simulink-r2025b.desktop` entry executable:

```
sudo chmod +x /usr/share/applications/simulink-r2025b.desktop
```

2.2) Create a Simulink Launcher File

Type the following (mkdir the bin directory if it does not exist). This creates an empty `simulink_open.sh` file

```
nano /usr/local/bin/simulink_open.sh
```

Paste the following in that file

```
#!/bin/bash

FILE="$1"

if [ -z "$FILE" ]; then
    # No file provided → just open Simulink
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "simulink"
else
    # File provided → extract directory and base name
    DIR="$(dirname "$FILE")"
    BASE="$(basename "$FILE")"
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "cd('$DIR');
open_system('$BASE');"
fi
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab_open.sh` file executable:

```
chmod +x /usr/local/bin/matlab_open.sh
```

When done, you need to refresh the applications database. To do so, either type the following:

```
sudo update-desktop-database
```

Or simply log out and log back in.

Now you should see **Simulink R2025b** in your Applications menu with an icon.

2.3) Create a MATLAB Icon

Same as for MATLAB, create a dedicated icon, then place it to the proper folder as follows:

```
sudo mv simulink.png /usr/local/MATLAB/R2025b/extern/icons/
```

2.0) Installing Packages

For WingLoop use, the following MATLAB Add-Ons need to be installed:

- MATLAB Compiler
- Simulink Compiler
- FMU Builder for Simulink

It is possible that MATLAB creates issues when attempting to install MathWorks Add-Ons. If so, for the case of the **Simulink Compiler** and **Matlab Compiler** packages, it is possible that one needs to activate the C++ libraries:

```
mex -setup
mex -setup cpp
```

It is also possible that matlab blocks the package installation if it is not opened as sudo. If this event happens, you have two options:

- Either you run the installer again, selecting the appropriate packages (same commands as above, copied here for sake of use)

```
xhost +SI:localuser:root
sudo -H ./install
xhost -SI:localuser:root
```

- Or open MATLAB as sudo: sudo

```
/usr/local/MATLAB/R2025b/bin/matlab
```